

Program : Diploma in Textile Technology	
Course Code : 5068	Course Title: Textile Colouration and Finishing Lab
Semester : 5	Credits: 4
Course Category: Program core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 60

Course Objectives:

- This course aims to impart practical knowledge on preparation and colouration process on textiles and the basics of printing technology by which the decorative designs can be produced on the textiles.

Course Prerequisites:

Topic	Course code	Course name	Semester
Processing and Dyeing of cotton using water soluble and insoluble dyes		Textile Wet Processing Practical-1	3
Processing and Dyeing of cotton using water soluble and		Textile Wet processing	3
Dyeing of silk, wool and polyester using various dyes and finishing process. Introduction to various printing process		Textile Colouration & Finishing	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Demonstrate the colouring of wool and silk material by using water soluble dyes.	12	Applying
CO2	Implement the colouring process on polyester material by using disperse dyes.	6	Applying
CO3	Execute a colouring process to produce a colour on blended material by using suitable dyes.	12	Applying
CO4	To operate different types of printing on fabrics	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3					
CO3			3				
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the colouring of wool and silk material by using water soluble dyes.		
M1.01	Dyeing of wool with acid dye.	3	Applying
M1.02	Dyeing of wool with basic dye.	3	Applying
M1.03	Dyeing of silk with acid dye.	3	Applying
M1.04	Dyeing of silk with basic dye.	3	Applying
CO2	Implement the colouring process on polyester material by using disperse dyes..		
M2.01	Dyeing of polyester material with disperse dye- carrier method & Analyse fastness	3	Applying
M2.02	Dyeing of polyester material with disperse dye- HTHP method	3	Applying
	Lab Exam - I	1.5	
CO3	Execute a colouring process to produce a colour on blended material by using suitable dyes.		
M3.01	Dyeing of Polyester/Cotton material with Disperse and Reactive Dyes.	3	Applying
M3.02	Dyeing of Polyester/Cotton material with Disperse and Vat Dyes.	3	Applying
M3.03	Compare the shade efficiency of Polyester/Cotton material with Disperse and Reactive Dyes , Polyester/Cotton material with Disperse and Vat Dyes using colour matching cabinet and choose the dye combination.	3	Applying
M3.04	Dyeing of Polyester/Viscose material with Disperse and Vat Dyes.	3	Applying

CO4	To operate different types of printing on fabrics		
M4.01	Printing of cotton cloth using blocks with Direct dyes	2	Applying
M4.02	Printing of cotton cloth using Stencil with Reactive dyes	2	Applying
M4.03	Printing the cotton cloth using screens with Pigment colours	2	Applying
M4.04	Developing the designs using Batik style in mono/ multi colours using natural dye and test the washing fastness.	3	Applying
M4.05	Developing the designs using tie and dye to produce mono/multi colours using natural dyes	3	Applying
	Lab Exam - II	1.5	

The students should keep record for Textile wet processing II with samples and submit certified & bonafide record at the time of practical examination.

Text / Reference:

T/R	Book Title/Author
T1	V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan publishers Ahmedabad.
R2	E.R Trotman - Dyeing & Chemical Technology of textiles
R3	R.S.Prayag - Dyeing of polyester and polyester blends
R4	R.R.Chakravarthy - A glimpses on the Chemical Technology of Textile Fibres
R5	D.G.Kale - Principles of Textile Printing

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/116/102/116102016/
2	https://swayam.gov.in/
3	http://eacharya.inflibnet.ac.in/vidya-mitra/
4	www.youtube.com